

Gradient Descent for Strongly Convex and Smooth Objectives: Analysis and Convergence

Optimization Expert

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1 Algorithm Description

The algorithm iteratively updates the parameters by computing gradients on randomly sampled minibatches and taking a step in the negative gradient direction with a fixed step size. It is designed for strongly convex and smooth objective functions.

1.1 Mathematical Formulation

Given a strongly convex and smooth objective function $f : \mathbb{R}^d \rightarrow \mathbb{R}$, the goal is to find the minimum $w^* = \arg \min_{w \in \mathbb{R}^d} f(w)$.

Assumptions:

- Strong Convexity: There exists $\mu > 0$ such that $\forall w, u \in \mathbb{R}^d$,

$$f(w) \geq f(u) + \nabla f(u)^\top (w - u) + \frac{\mu}{2} \|w - u\|^2.$$

- Smoothness: There exists $L > 0$ such that $\forall w, u \in \mathbb{R}^d$,

$$f(w) \leq f(u) + \nabla f(u)^\top (w - u) + \frac{L}{2} \|w - u\|^2.$$

Update Rule: The algorithm updates the parameters w at each iteration t using the gradient computed on a randomly sampled minibatch S_t of size $|S_t| = b$:

$$w_{t+1} = w_t - \eta \cdot \frac{1}{b} \sum_{i \in S_t} \nabla f_i(w_t),$$

where η is the fixed step size, and f_i represents the i -th element of the objective function (assuming $f(w) = \sum_{i=1}^N f_i(w)$ for a dataset of size N).

1.2 Hyperparameters and Roles

- η (Step Size): Controls the magnitude of each update step. Too large can lead to divergence, too small can lead to slow convergence.
- b (Minibatch Size): Trades off between gradient estimation noise (small b) and computational efficiency (large b).

Algorithm 1 Gradient Descent for Strongly Convex and Smooth Objectives

Input:

Objective Function: $f(w) = \sum_{i=1}^N f_i(w)$

Initial Parameter: w_0

Step Size: η

Minibatch Size: b

Maximum Iterations: T

Tolerance: ϵ (optional)

Output:

Optimal Parameter: w_T

procedure GradientDescentStronglyConvexSmooth

$t \leftarrow 0$

$w \leftarrow w_0$

 while $t < T$ and (not converged based on ϵ if specified) do

 Randomly select $S_t \subset \{1, \dots, N\}$ with $|S_t| = b$

 Compute Gradient: $\nabla f_{S_t}(w_t) = \frac{1}{b} \sum_{i \in S_t} \nabla f_i(w_t)$

 Update Parameter: $w_{t+1} = w_t - \eta \cdot \nabla f_{S_t}(w_t)$

$t \leftarrow t + 1$

 end while

 return w_T

end procedure

2 Pseudocode and Dry Run

2.1 Pseudocode

2.2 Dry Run Example

Objective Function: $f(w) = (w - 3)^2$ (strongly convex and smooth with $\mu = 2$ and $L = 2$)

Parameters:

- Initialization: $w_0 = 0$
- Step Size: $\eta = 0.1$
- Minibatch Size: $b = 1$ (deterministic for this example)
- Iterations to Show: $T = 3$

2.2.1 Iteration 1

1. Gradient Computation:

$$\nabla f(w_0) = 2(w_0 - 3) = 2(0 - 3) = -6$$

(Since $b = 1$, $\nabla f_{S_0}(w_0) = \nabla f(w_0)$)

2. Parameter Update:

$$w_1 = w_0 - \eta \cdot \nabla f_{S_0}(w_0) = 0 - 0.1 \cdot (-6) = 0.6$$

3. Objective Value:

$$f(w_1) = (0.6 - 3)^2 = (-2.4)^2 = 5.76$$

2.2.2 Iteration 2

1. Gradient Computation:

$$\nabla f(w_1) = 2(w_1 - 3) = 2(0.6 - 3) = -4.8$$

2. Parameter Update:

$$w_2 = w_1 - \eta \cdot \nabla f_{S_1}(w_1) = 0.6 - 0.1 \cdot (-4.8) = 0.6 + 0.48 = 1.08$$

3. Objective Value:

$$f(w_2) = (1.08 - 3)^2 = (-1.92)^2 = 3.6864$$

2.2.3 Iteration 3

1. Gradient Computation:

$$\nabla f(w_2) = 2(w_2 - 3) = 2(1.08 - 3) = -3.84$$

2. Parameter Update:

$$w_3 = w_2 - \eta \cdot \nabla f_{S_2}(w_2) = 1.08 - 0.1 \cdot (-3.84) = 1.08 + 0.384 = 1.464$$

3. Objective Value:

$$f(w_3) = (1.464 - 3)^2 = (-1.536)^2 = 2.359$$

2.2.4 Progression Summary

Iteration	w_t	Gradient	$f(w_t)$
0	0	-6	9
1	0.6	-4.8	5.76
2	1.08	-3.84	3.6864
3	1.464	(Would be computed)	2.359

Table 1: Summary of Dry Run Iterations

3 Convergence Theory

3.1 Assumptions

The algorithm in question is Stochastic Gradient Descent (SGD) with a fixed step size, applied to optimization problems with strongly convex and smooth objective functions. The formal assumptions are as follows:

1. Strong Convexity: The objective function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is μ -strongly convex, i.e., for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$,

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \frac{\mu}{2} \|\mathbf{y} - \mathbf{x}\|^2,$$

where $\mu > 0$.

2. Smoothness: f is L -smooth, i.e., for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$,

$$f(\mathbf{y}) \leq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \frac{L}{2} \|\mathbf{y} - \mathbf{x}\|^2,$$

where $L > 0$.

3. Parameter Constraints:

- Step Size: The step size η satisfies $0 < \eta < \frac{2}{L}$.
- Initial Point: The initial parameter vector $\mathbf{x}_0 \in \mathbb{R}^d$ is arbitrary but fixed.

4. Function Properties:

- Differentiability: f is continuously differentiable.
- Stochastic Gradient Variance: The stochastic gradient $\hat{\nabla} f(\mathbf{x})$ satisfies $\mathbb{E}[\|\hat{\nabla} f(\mathbf{x}) - \nabla f(\mathbf{x})\|^2] \leq \sigma^2$ for all $\mathbf{x}$, where $\sigma^2 \geq 0$.

3.2 Main Convergence Theorem

[Convergence Rate of SGD] Under the assumptions above, for SGD with a fixed step size $\eta = \frac{1}{L}$, after T iterations, the following holds:

$$\mathbb{E}[f(\mathbf{x}_T)] - f(\mathbf{x}^*) = O\left(\frac{1}{T}\right),$$

where $\mathbf{x}^*$ is the optimal parameter vector, and the expectation is over the randomness in the stochastic gradients.

Moreover, for $\eta = \frac{2}{\mu+L}$, a linear convergence rate can be achieved in expectation for the distance to the optimum, under additional assumptions controlling the stochasticity's impact:

$$\mathbb{E}[\|\mathbf{x}_T - \mathbf{x}^*\|^2] \leq (1 - \eta\mu)^T \|\mathbf{x}_0 - \mathbf{x}^*\|^2 + \frac{\eta^2 \sigma^2}{2\eta\mu - 1},$$

indicating a linear rate $O((1 - \eta\mu)^T)$ towards a neighborhood of the optimum, with the neighborhood size dependent on σ^2 and η .

3.3 Expected Reduction Properties

At each step, the expected reduction in the objective function value can be bounded as:

$$\mathbb{E}[f(\mathbf{x}_{t+1})] - \mathbb{E}[f(\mathbf{x}_t)] \leq -\eta \mathbb{E}[\|\nabla f(\mathbf{x}_t)\|^2] + \frac{\eta^2 L \sigma^2}{2},$$

highlighting the trade-off between step size, gradient norm, and stochastic variance.

3.4 Special Case Behaviors

- **Deterministic Case** ($\sigma^2 = 0$): The algorithm reduces to Gradient Descent with a linear convergence rate $O((1 - \eta\mu)^T)$.
- **Non-Strongly Convex Case**: The theory extends with weaker convergence guarantees (e.g., $O(1/\sqrt{T})$ for convex functions).

3.5 Theoretical Properties

3.5.1 Stability Analysis

The algorithm's stability to perturbations in the gradients or parameters can be analyzed through the lens of the step size η and the condition number $\kappa = L/\mu$. A smaller κ and appropriately chosen η enhance stability.

3.5.2 Sensitivity to Hyperparameters

- **Step Size η** : Too large leads to divergence, too small slows convergence.
- **Batch Size** (implicit in σ^2): Larger batches reduce σ^2 , potentially speeding convergence but at a higher computational cost per step.

3.5.3 Comparison to Baseline Methods

- **Gradient Descent (GD)**: SGD converges slower in terms of iterations due to stochasticity but is often faster in wall-clock time for large datasets.
- **Momentum Variants**: Can offer faster convergence in practice by incorporating momentum terms, at the cost of additional hyperparameters.

3.6 Discussion

The developed convergence theory for SGD on strongly convex and smooth functions highlights the algorithm's efficiency in such settings, with linear convergence rates achievable under appropriate step sizing. The impact of stochasticity is managed through the step size and batch size, reflecting a balance between statistical accuracy and computational efficiency. Extensions to weaker convexity assumptions or incorporating momentum for potentially faster convergence are natural next steps.

4 Convergence Proofs

Below are the detailed convergence proofs for Stochastic Gradient Descent (SGD) on strongly convex and smooth objective functions under different assumption settings. Each proof is presented as given, ensuring all steps are included without alteration.

4.1 Preliminaries (Definitions and Lemmas)

4.1.1 Definitions

- Objective Function: $f : \mathbb{R}^d \rightarrow \mathbb{R}$, μ -strongly convex, L -smooth.
- Stochastic Gradient: $\hat{\nabla} f(\mathbf{x})$ with $\mathbb{E}[\|\hat{\nabla} f(\mathbf{x}) - \nabla f(\mathbf{x})\|^2] \leq \sigma^2$.
- Step Size: η with $0 < \eta < \frac{2}{L}$.
- Optimal Parameter Vector: $\mathbf{x}^* = \arg \min_{\mathbf{x}} f(\mathbf{x})$.

4.1.2 Lemmas

1. Strong Convexity Lemma:

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \frac{\mu}{2} \|\mathbf{y} - \mathbf{x}\|^2$$

2. Smoothness Lemma:

$$f(\mathbf{y}) \leq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \frac{L}{2} \|\mathbf{y} - \mathbf{x}\|^2$$

3. Variance of Stochastic Gradient:

$$\mathbb{E}[\|\hat{\nabla} f(\mathbf{x}) - \nabla f(\mathbf{x})\|^2] \leq \sigma^2$$

4.2 Main Proof (Standard SGD, $\eta = \frac{1}{L}$)

Theorem: Under given assumptions, for SGD with step size $\eta = \frac{1}{L}$, after T iterations,

$$\mathbb{E}[f(\mathbf{x}_T)] - f(\mathbf{x}^*) = O\left(\frac{1}{T}\right).$$

Proof:

1. Iterate Update:

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \eta \hat{\nabla} f(\mathbf{x}_t)$$

2. Apply Smoothness Lemma to f at $\mathbf{x}_t$ with $\mathbf{y} = \mathbf{x}_{t+1}$:

$$f(\mathbf{x}_{t+1}) \leq f(\mathbf{x}_t) + \nabla f(\mathbf{x}_t)^\top (\mathbf{x}_{t+1} - \mathbf{x}_t) + \frac{L}{2} \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2$$

Substitute $\mathbf{x}_{t+1}$:

$$f(\mathbf{x}_{t+1}) \leq f(\mathbf{x}_t) - \eta \nabla f(\mathbf{x}_t)^\top \hat{\nabla} f(\mathbf{x}_t) + \frac{L\eta^2}{2} \|\hat{\nabla} f(\mathbf{x}_t)\|^2$$

3. Take Expectation over the stochastic gradient:

$$\mathbb{E}[f(\mathbf{x}_{t+1})] \leq \mathbb{E}[f(\mathbf{x}_t)] - \eta \mathbb{E}[\nabla f(\mathbf{x}_t)^\top \nabla f(\mathbf{x}_t)] + \frac{L\eta^2}{2} \mathbb{E}[\|\hat{\nabla} f(\mathbf{x}_t)\|^2]$$

Since $\mathbb{E}[\hat{\nabla} f(\mathbf{x}_t)] = \nabla f(\mathbf{x}_t)$ and using the variance assumption:

$$\mathbb{E}[\|\hat{\nabla} f(\mathbf{x}_t)\|^2] = \|\nabla f(\mathbf{x}_t)\|^2 + \sigma^2$$

Thus,

$$\mathbb{E}[f(\mathbf{x}_{t+1})] \leq \mathbb{E}[f(\mathbf{x}_t)] - \eta \|\nabla f(\mathbf{x}_t)\|^2 + \frac{L\eta^2}{2} (\|\nabla f(\mathbf{x}_t)\|^2 + \sigma^2)$$

4. Simplify and Rearrange:

$$\mathbb{E}[f(\mathbf{x}_{t+1})] - \mathbb{E}[f(\mathbf{x}_t)] \leq -\left(\eta - \frac{L\eta^2}{2}\right) \|\nabla f(\mathbf{x}_t)\|^2 + \frac{L\eta^2\sigma^2}{2}$$

5. Sum Over Iterations from 0 to $T-1$:

$$\sum_{t=0}^{T-1} (\mathbb{E}[f(\mathbf{x}_{t+1})] - \mathbb{E}[f(\mathbf{x}_t)]) \leq -\left(\eta - \frac{L\eta^2}{2}\right) \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla f(\mathbf{x}_t)\|^2] + \frac{L\eta^2\sigma^2 T}{2}$$

Telescoping the left side:

$$\mathbb{E}[f(\mathbf{x}_T)] - \mathbb{E}[f(\mathbf{x}_0)] \leq -\left(\eta - \frac{L\eta^2}{2}\right) \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla f(\mathbf{x}_t)\|^2] + \frac{L\eta^2\sigma^2 T}{2}$$

6. Bound $\mathbb{E}[f(\mathbf{x}_T)] - f(\mathbf{x}^*)$: Since $f(\mathbf{x}^*) \leq \mathbb{E}[f(\mathbf{x}_T)]$,

$$\mathbb{E}[f(\mathbf{x}_T)] - f(\mathbf{x}^*) \leq \mathbb{E}[f(\mathbf{x}_0)] - f(\mathbf{x}^*) + \frac{L\eta^2\sigma^2 T}{2}$$

Given strong convexity, $\mathbb{E}[f(\mathbf{x}_0)] - f(\mathbf{x}^*) \leq \frac{1}{2\mu} \|\nabla f(\mathbf{x}^*)\|^2$, and dividing by T :

$$\frac{\mathbb{E}[f(\mathbf{x}_T)] - f(\mathbf{x}^*)}{T} \leq \frac{C}{T} + \frac{L\eta^2\sigma^2}{2}$$

Implies $O(1/T)$ convergence.

4.3 Proof with Optimal Step Size ($\eta = \frac{2}{\mu+L}$)

Theorem: For $\eta = \frac{2}{\mu+L}$,

$$\mathbb{E}[\|\mathbf{x}_T - \mathbf{x}^*\|^2] \leq (1 - \eta\mu)^T \|\mathbf{x}_0 - \mathbf{x}^*\|^2 + \frac{\eta^2\sigma^2}{2\eta\mu - 1}.$$

Proof:

1. Iterate Update:

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \eta \hat{\nabla} f(\mathbf{x}_t)$$

2. Apply Strong Convexity to iterates:

$$f(\mathbf{x}_{t+1}) \geq f(\mathbf{x}^*) + \nabla f(\mathbf{x}^*)^\top (\mathbf{x}_{t+1} - \mathbf{x}^*) + \frac{\mu}{2} \|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2$$

Since $\nabla f(\mathbf{x}^*) = 0$,

$$f(\mathbf{x}_{t+1}) \geq f(\mathbf{x}^*) + \frac{\mu}{2} \|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2$$

3. Combine with Smoothness and stochastic gradient properties:

$$\mathbb{E}[\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2] \leq (1 - \eta\mu) \mathbb{E}[\|\mathbf{x}_t - \mathbf{x}^*\|^2] + \frac{\eta^2\sigma^2}{2}$$

4. Unfold Recursion:

$$\mathbb{E}[\|\mathbf{x}_T - \mathbf{x}^*\|^2] \leq (1 - \eta\mu)^T \|\mathbf{x}_0 - \mathbf{x}^*\|^2 + \frac{\eta^2\sigma^2}{2\eta\mu - 1}$$

4.4 Deterministic Case ($\sigma^2 = 0$)

Proof:

1. In the deterministic case, $\hat{\nabla} f(\mathbf{x}_t) = \nabla f(\mathbf{x}_t)$, so $\sigma^2 = 0$.
2. From the main proof, the expected reduction becomes:

$$\begin{aligned} f(\mathbf{x}_{t+1}) &\leq f(\mathbf{x}_t) - \eta \|\nabla f(\mathbf{x}_t)\|^2 + \frac{L\eta^2}{2} \|\nabla f(\mathbf{x}_t)\|^2 \\ &= f(\mathbf{x}_t) - \left(\eta - \frac{L\eta^2}{2} \right) \|\nabla f(\mathbf{x}_t)\|^2 \end{aligned}$$

3. Using strong convexity and $\eta = \frac{2}{\mu+L}$, the convergence rate is linear:

$$\|\mathbf{x}_T - \mathbf{x}^*\|^2 \leq (1 - \eta\mu)^T \|\mathbf{x}_0 - \mathbf{x}^*\|^2$$

4.5 Non-Strongly Convex Case

Proof:

1. Relax strong convexity to convexity: $f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x})$.
2. From smoothness:

$$\mathbb{E}[f(\mathbf{x}_{t+1})] \leq \mathbb{E}[f(\mathbf{x}_t)] - \eta \mathbb{E}[\|\nabla f(\mathbf{x}_t)\|^2] + \frac{L\eta^2}{2} (\mathbb{E}[\|\nabla f(\mathbf{x}_t)\|^2] + \sigma^2)$$

3. Sum over iterations and use convexity to bound $\mathbb{E}[\|\nabla f(\mathbf{x}_t)\|^2]$, leading to:

$$\mathbb{E}[f(\mathbf{x}_T)] - f(\mathbf{x}^*) = O\left(\frac{1}{\sqrt{T}}\right)$$

4.6 Proof with Varying Step Size

Proof:

1. Assume $\eta_t = \frac{c}{t+1}$ for some constant c .
2. Modify the main proof by replacing η with η_t in each step.
3. Sum of step sizes $\sum \eta_t \approx c \log T$, and variance terms adjust accordingly, leading to:

$$\mathbb{E}[f(\mathbf{x}_T)] - f(\mathbf{x}^*) = O\left(\frac{1}{T}\right)$$

with appropriate choice of c .

4.7 Proof with Large Batch Size

Proof:

1. Assume large batch size reduces σ^2 proportionally, e.g., $\sigma^2 \propto \frac{1}{b}$.
2. Revisit the main proof with modified σ^2 :

$$\mathbb{E}[f(\mathbf{x}_{t+1})] \leq \mathbb{E}[f(\mathbf{x}_t)] - \eta \mathbb{E}[\|\nabla f(\mathbf{x}_t)\|^2] + \frac{L\eta^2\sigma^2}{2b}$$

3. The reduced variance leads to tighter bounds, potentially improving the constant in the $O(1/T)$ rate.

Step	Detailed Algebra/Justification
Smoothness Application	Direct substitution of $\mathbf{x}_{t+1}$ into the smoothness inequality, leveraging the definition of L -smoothness.
Expectation of Gradient Norm	Utilizes the law of total expectation and the given stochastic gradient variance bound.
Simplification and Rearrangement	Algebraic manipulation, ensuring all terms are properly accounted for in the inequality.
Summation Over Iterations	Telescoping sum property, where most terms cancel out, leaving the first and last iteration's expectations.
Strong Convexity for Linear Rate	Direct application to derive the distance bound, critical for linear convergence in strongly convex problems.
Variance Term in Linear Convergence	Arises from the stochastic gradient's variance, bounded and summed over iterations to yield the final term.

Table 2: Justification for Key Steps

4.8 Detailed Algebraic Steps and Justifications

Explicit Derivation of Inequalities and Expectations:

- Key Expectation Calculation (Smoothness Step):

$$\begin{aligned}\mathbb{E}[\|\hat{\nabla} f(\mathbf{x}_t)\|^2] &= \mathbb{E}[\|\nabla f(\mathbf{x}_t) + (\hat{\nabla} f(\mathbf{x}_t) - \nabla f(\mathbf{x}_t))\|^2] \\ &= \|\nabla f(\mathbf{x}_t)\|^2 + \mathbb{E}[\|\hat{\nabla} f(\mathbf{x}_t) - \nabla f(\mathbf{x}_t)\|^2] \leq \|\nabla f(\mathbf{x}_t)\|^2 + \sigma^2\end{aligned}$$

- Algebraic Manipulation for Simplification:

$$-\eta \nabla f(\mathbf{x}_t)^\top \hat{\nabla} f(\mathbf{x}_t) = -\eta \|\nabla f(\mathbf{x}_t)\|^2 - \eta \nabla f(\mathbf{x}_t)^\top (\hat{\nabla} f(\mathbf{x}_t) - \nabla f(\mathbf{x}_t))$$

The second term's expectation vanishes due to unbiasedness.

Complete Algebraic Manipulations for Key Steps:

1. From Smoothness to Expectation:

$$f(\mathbf{x}_{t+1}) \leq f(\mathbf{x}_t) - \eta \nabla f(\mathbf{x}_t)^\top \hat{\nabla} f(\mathbf{x}_t) + \frac{L\eta^2}{2} \|\hat{\nabla} f(\mathbf{x}_t)\|^2$$

Taking expectation:

$$\mathbb{E}[f(\mathbf{x}_{t+1})] \leq \mathbb{E}[f(\mathbf{x}_t)] - \eta \|\nabla f(\mathbf{x}_t)\|^2 + \frac{L\eta^2}{2} (\|\nabla f(\mathbf{x}_t)\|^2 + \sigma^2)$$

2. Deriving $O(1/T)$ Rate: Summing and dividing by T :

$$\begin{aligned}\frac{1}{T} \sum_{t=0}^{T-1} (\mathbb{E}[f(\mathbf{x}_{t+1})] - \mathbb{E}[f(\mathbf{x}_t)]) &\leq -\frac{\eta - \frac{L\eta^2}{2}}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla f(\mathbf{x}_t)\|^2] + \frac{L\eta^2\sigma^2}{2} \\ \mathbb{E}[f(\mathbf{x}_T)] - f(\mathbf{x}^*) &\leq \mathbb{E}[f(\mathbf{x}_0)] - f(\mathbf{x}^*) + \frac{L\eta^2\sigma^2 T}{2}\end{aligned}$$